

Physics-informed neural networks for hydraulic transient analysis in pipeline systems

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ABSTRACT

In water pipeline systems, monitoring and predicting hydraulic transient events are important to ensure the proper operation of pressure control devices (e.g., pressure reducing valves) and prevent potential damages to the network infrastructure. Simulating transient pressures using traditional numerical methods, however, require a complete model with known boundary and initial conditions, which is rarely able to obtain in a real system. This paper proposes a new physics-based and data-driven method for targeted transient pressure reconstruction without the need of having a complete pipe system model. The new method formulates a physics-informed neural network (PINN) by incorporating both measured data and physical laws of the transient flow in the training process. This enables the PINN to learn and explore hidden information of the hydraulic transient (e.g., boundary conditions and wave damping characteristics) that is embedded in the measured data. The trained PINN can then be used to predict transient pressures at any location of the pipeline. Results from two numerical and one experimental case studies showed a high accuracy of the pressure reconstruction using the proposed approach. In addition, a series of sensitivity analyses have been conducted to determine the optimal hyperparameters in the PINN and to understand the effects of the sensor configuration on the model performance.

1. Introduction

Hydraulic transient phenomena widely exist in pipeline systems, from water distribution networks (Boulos et al., 2005) and hydroelectric power plants (Afshar et al., 2010) to gas pipelines (Reddy et al., 2006) or industrial piping systems. The pressure induced by hydraulic transient events (e.g., valve closure) can change rapidly and reach an extremely level (El-Hazek et al., 2020). Those extreme pressures have significant impacts on the reliability of a pipeline system and may result in device failures, pipe bursts (Misiunas et al., 2006), water loss and contaminant intrusion (Ebacher et al., 2012).

The adoption of smart sensor networks, with a number of high-frequency loggers deployed across the water pipeline system (Xing and Sela, 2019; Wu et al., 2021), can be used to capture and monitor the transient pressure. Large amounts of collected data enable smart water network operation and management (PUB, 2016) to mitigate the negative impact of transient events. However, high-frequency loggers are often sparsely distributed in a water distribution system (WDS), leading to limited monitoring of the pressure variations among the entire system. Locations without measurement devices may possess a hidden risk

since abnormal pressures occurring in these regions could not be recorded by existing sensors at other locations. Therefore, transient analysis for unmonitored locations using pressure measurements at sparsely existing loggers is crucial for water utilities to better understand the impact of transient events and thus take preventive operational actions.

Over the past several decades, various numerical models have been developed to simulate and predict pressure variations of hydraulic transient events, including the wave characteristic method (Wood et al., 1966), method of characteristics (MOC) (Wylie and Streeter, 1993), finite-difference method (FDM) (Chaudhry, 2014), and finite volume method (FVM) (Hwang and Chung, 2002). These physical-based methods construct the transient pressure by solving a set of partial differential equations (PDEs) that govern the transient flow. However, the accuracy and reliability of these methods are built on full knowledge of the boundary conditions and transient excitation, which is very challenging to obtain in a real system. In practice, the hydraulic devices as boundary nodes in these models are difficult to formulate accurately (e.g., how a valve is closed is normally unknown). Furthermore, the physical characteristics of the pressure wave propagation involve

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various uncertain factors that are normally difficult to calibrate in the field. These include but are not limited to pipe network parameters (e.g., pipe diameters, wave speeds), the dynamic effect of unsteady friction (Brunone et al., 2000; Vardy and Brown, 1995; Vitkovský et al., 2000) and pipe-wall viscoelasticity (Covas et al., 2005).

With the increasing availability of wireless sensors in recent years, data-driven solvers become popular in WDS modeling and analysis since they can avoid the requirement of a detailed physical model. These approaches use measurement data, such as pressures and flowrates, to infer the hydraulic states and system parameters. Applications of data-driven solvers in pipeline systems can be found in various research topics, ranging from pipe condition assessment (Khan et al., 2006; Jafar et al., 2010) and water quality evaluation (Ömer Faruk, 2010; Khatri et al., 2019) to pipe failure detection (Wu et al., 2016; Zhou et al., 2019; Bohorquez et al., 2021). The data-driven solvers, especially machine learning, have also been applied for pressure prediction at steady state (Lima et al., 2017). For example, Ridolfi et al. (2014) applied the artificial neural network (ANN) with an entropy-based method to determine water pressure at unknown nodes. The small discrepancy between simulated and observed water pressure at each node demonstrated the capability of this method for pressure prediction. Hajgató et al. (2021) used a graph neural network (GNN), a machine learning model that can explicitly include the topology information of a pipe network, to reconstruct all the nodal pressures based on partially observed nodal pressures in three different WDSs. A GNN model was also proposed by Xing and Sela (2022) to estimate the flows and heads in WDSs. In this study, the physical laws that govern the steady-state hydraulics were incorporated into the training process to improve the performance of the trained model.

Although the above examples achieved some success in pressure and flowrate prediction in hydraulic steady states, there is a lack of study on transient state prediction, specifically for predicting transient pressure in pipeline networks. The key obstacles to this problem are, firstly, the sparsely measured data due to a limited number of sensors and large uncertainties that exist in practice, such as uncertain parameters of pipes, dispersion and dissipation of pressure waves (also referred to as damping) and background noise of measured data. Secondly, it can be seen that the traditional physics-based methods for transient pressure prediction are not well fit to use with field measurement data as they require complete knowledge of the transient source/excitation to characterize the acceleration/deceleration of transient flows (i.e., solving a set of complex PDEs).

The physics-informed neural network (PINN) proposed by Raissi et al. (2019) is a potential effective tool to overcome these obstacles. Different to purely data-driven models, the PINN can utilize hidden physical information from the observed dataset to support solving nonlinear PDEs. Given its high accuracy in predicting complex dynamics from incomplete models and sparse data, the PINN has recently been applied to various fields, such as computational fluid dynamics (CFD) (Buhendwa et al., 2021; Mao et al., 2020; Rao et al., 2020), power systems (Misyr et al., 2020), metamaterial design (Chen et al., 2020) and transient electrodynamics (Noakoosteen et al., 2020).

In this paper, a PINN-based transient analysis method has been proposed for transient pressure prediction. To the authors' knowledge, this is the first such attempt to enhance hydraulic transient analysis without a complete model by incorporating a PINN into the analysis. Compared with the traditional physics-based numerical methods, the PINN-based method developed in this paper incorporates the hidden information (physical characteristics such as wave damping and boundary conditions, etc) embedded in the measured data into the transient analysis and thus does not require a fully complete physics-based model. Compared with the purely data-driven methods, the PINN-based method incorporates the physical laws of transient flow into the training process and thus provides physically consistent solutions even with sparsely measured data (Karniadakis et al., 2021).

2. Methodology

For the purpose of developing the PINN-based hydraulic transient analysis to determine the pressure and flowrate variations, it is necessary to understand the fundamental of hydraulic transient. In this section, the governing equations of hydraulic transients are, therefore, firstly depicted. Then, the PINN architecture is illustrated, which incorporates both the above-mentioned physical laws (governing equations) and available measurements.

2.1. Mathematical representation of hydraulic transients

Dynamic behaviours of the transient flow in pressure pipes are represented by a set of conservation of mass and momentum equations. These are a set of PDEs since the flowrate $q(x, t)$ and pressure (i.e., piezometric head) $h(x, t)$ are functions of distance (x) and time (t). Assuming the flow is one-dimensional and slightly compressible in a pressurized pipe having linearly elastic walls, the governing equations for the transient flow can be written as follows (Wylie and Streeter, 1993):

$$A \frac{\partial q}{\partial t} + q \frac{\partial q}{\partial x} + g A^2 \frac{\partial h}{\partial x} + f \frac{|q|q}{2D} = 0 \quad (1)$$

$$A \frac{\partial h}{\partial t} + q \frac{\partial h}{\partial x} + \frac{a^2}{g} \frac{\partial q}{\partial x} = 0 \quad (2)$$

where $h(x, t)$ is the piezometric head, $q(x, t)$ is the flowrate; x is the distance along the pipeline and t is time; a is the wave speed, g is the gravitational acceleration, D is the pipe diameter and $A = \pi D^2/4$ is the cross-section area of the pipeline and f is the Darcy–Weisbach friction factor.

2.2. Problem definition for the PINN-based hydraulic transient analysis

For a pipeline system with unknown boundary conditions, as shown in Fig. 1, it is assumed that the pipe parameters $\lambda = \{a, f, D, L\}$ are given and L is the pipe length. The system is characterized by a state vector Φ that consists of hydraulic head and flowrate and depends on time and space. This state vector Φ can be expressed by:

$$\Phi(x, t) = [h(x, t), q(x, t)]x \in [0, L], t \in [0, T] \quad (3)$$

where T is the total time of the investigated transient event.

To solve the system state Φ with unknown boundaries, it is assumed that there are k_d sensors installed on the pipe wall at locations $P_1, P_2, \dots, P_{k_d}$ to collect additional information. These locations are referred to as observation points as shown in Fig. 1. The subscript d indicates the observed data.

Define M_d as the set of all the spatially-temporally observed data points and M_d^i as the i_{th} observed data point consisting of the recorded physical quantity φ_d^i by the sensor and its spatial-temporal characteristics x_d^i and t_d^i . The term φ_d^i can be either the observed pressure head h or the observed flowrate q , determined by the type of sensors. Each observation contains N data points in time series. Therefore, the set of all the observed data points can be expressed as:

$$M_d = \left\{ \overbrace{M_d^1, M_d^2, \dots, M_d^N}^{P_1}, \overbrace{M_d^{N+1}, \dots, M_d^{2N}}^{P_2}, \dots, \overbrace{M_d^{(k-1)N+1}, \dots, M_d^{N_d}}^{P_{k_d}} \right\} \quad (4)$$

$$= \{x_d^i, t_d^i, \varphi_d^i\}_{i=1}^{N_d}, \varphi_d^i \in \Phi(x_d^i, t_d^i)$$

where $N_d = k_d \times N$ and is the total number of observed data points at all the observation points and at all the time steps.

Using this definition, the hydraulic transient analysis can be taken as the estimation of the system states Φ (outputs) at any space x and time t

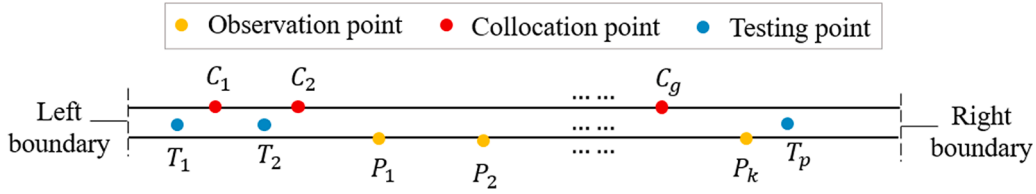


Fig. 1. A pipeline system with unknown left and right boundaries. At observation points, pressure or flowrate traces are measured as observation data. At collocation points, physical laws are imposed by Eqs. (1) and (2). At testing points, the pressure and flowrate traces are predicted and evaluated to validate the accuracy and efficacy of the PINN approach's solutions.

(inputs) when the system parameters λ and observation dataset $\mathbf{M}_d$ are available. Without knowing the initial and boundary conditions, the Eqs. (1) and (2) cannot be solved. In other words, the state variables Φ cannot be obtained by conventional numerical transient solvers that use this set of transient governing equations. In addition, the inevitable uncertainties existing in the pipeline system and observation data may affect the accuracy of the result from these deterministic methods where all the parameters and data require to be known exactly. To solve this state estimation problem with various uncertainties, a PINN-based transient analysis method is formulated in the next section.

2.2. Physics-informed neural networks

To formulate a PINN for transient flow which is governed by the aforementioned PDEs (i.e., Eqs. (1) and (2)), a deep neural network (DNN) is first constructed as a surrogate of the PDE solution. Here, the feed-forward neural network (FNN) which is relatively simple but sufficient for most PDE problems is employed. The FNN is trained by a two-step process: forward propagation and back propagation. In the forward propagation, the FNN takes the spatial-temporal variables $\{x, t\}$ as input and the mix-variable solution $\hat{\Phi}(x, t; \theta) = [\hat{h}(x, t; \theta), \hat{q}(x, t; \theta)]$ as output. The symbol θ is a vector containing all weights and biases in the neural network that are adjusted by a gradient descent algorithm in the back propagation. The hat above the variables indicates that the variables are calculated by the neural network.

The next key step is to constrain the output $\hat{\Phi}(x, t; \theta)$ to satisfy the data observations as well as the physical laws defined by Eqs. (1) and (2). To quantify the compliance of the neural network with the physical laws, the residuals are defined as:

$$F_1 = A \frac{\partial \hat{q}}{\partial t} + \hat{q} \frac{\partial \hat{q}}{\partial x} + gA^2 \frac{\partial \hat{h}}{\partial x} + f \frac{|\hat{q}| \hat{q}}{2D} \quad (5)$$

$$F_2 = A \frac{\partial \hat{h}}{\partial t} + \hat{q} \frac{\partial \hat{h}}{\partial x} + \frac{a^2}{g} \frac{\partial \hat{q}}{\partial x} \quad (6)$$

The residual vector $\mathbf{F} = [F_1, F_2]$ represents the degree of compliance of the PINN with the physical laws, with small residual values indicating high compliance. The spatial and time derivatives $\frac{\partial \hat{h}}{\partial x}$, $\frac{\partial \hat{h}}{\partial t}$, $\frac{\partial \hat{q}}{\partial x}$ and $\frac{\partial \hat{q}}{\partial t}$ in F_1 and F_2 are computed by using the Automatic Differentiation (AD) algorithm (Baydin et al., 2018). Note that the AD can accurately calculate the derivative of numeric functions expressed as a graph structure, especially for neural networks, since it does not require any approximation schemes (e.g., the finite difference approximation) and avoids the errors such as truncation and round-off errors. Thus, the PINN is a mesh-free method without the requirement of a discrete mesh to solve the PDEs.

A set of predefined points (herein, referred to as collocation points) at k_f different locations $C_1, C_2, \dots, C_{k_f}$ (see Fig. 1) with N time steps for each are selected to evaluate their residuals. The collocation dataset can be defined as

$$\mathbf{M}_f = \left\{ x_f^j, t_f^j \right\}_{j=1}^{N_f} \quad (7)$$

where the subscript f indicates the collocation points, N_f indicates the number of the data points at all the collocation points and at all the time steps, and $N_f = k_f \times N$. Note that the solutions from the PINN at these points must comply with the partial differential equations and the evaluation of their residuals does not require the true 'target value' of Φ (i.e., the observation data). Therefore, these points can be placed at any location along the pipelines. Since it is impossible to make collocation points cover the whole continuous spatial-temporal domain, a limited number of collocation points are selected to account for the global PDEs. The number of collocation points can be determined by an Adaptive Collocation Method (van der Meer et al., 2021).

Different from other PINN applications, the PINN for transient flow analysis in this paper assumes the boundary and initial conditions are unknown for practical reasons. Based on the description above, the loss function $L(\theta)$ in the PINN is constructed by only considering terms corresponding to PDEs and the field observations, and is given by:

$$L(\theta) = L_{data} + w_f L_{PDE} \quad (8)$$

where w_f is the weighting value required to balance the PDE loss term L_{PDE} and the data loss term L_{data} , expressed by:

$$L_{data} = \frac{1}{N_d} \sum_{j=1}^{N_d} (\hat{\varphi}(x_d^j, t_d^j; \theta) - \varphi_d(x_d^j, t_d^j))^2 \quad (9)$$

$$L_{PDE} = \frac{1}{N_f} \sum_{j=1}^{N_f} (F_1^2(x_f^j, t_f^j; \theta) + F_2^2(x_f^j, t_f^j; \theta)) \quad (10)$$

In Eqs. (8)–(10), the data loss term L_{data} aims to fit the observed physical quantity φ_d (in Eq. (4)) with the corresponding $\hat{\varphi}$ from the neural network output; while the PDEs loss term L_{PDE} enforces the hydraulic transient flow structure imposed by Eqs. (1) and (2) at the collocation points by minimizing both the residuals F_1 and F_2 .

The architecture of the proposed PINNs for hydraulic transient flow is presented in Fig. 2. For the input layer, the input matrix is:

$$\text{Input} : \begin{bmatrix} x_d & t_d \\ x_f & t_f \end{bmatrix} = \begin{array}{cc} \text{Space} & \text{Time} \\ \left| \begin{array}{c} x_d^1 \\ x_d^2 \\ \dots \\ x_d^{N_d} \\ x_f^1 \\ x_f^2 \\ \dots \\ x_f^{N_f} \end{array} \right| & \left| \begin{array}{c} t_d^1 \\ t_d^2 \\ \dots \\ t_d^{N_d} \\ t_f^1 \\ t_f^2 \\ \dots \\ t_f^{N_f} \end{array} \right| \end{array} \quad (11)$$

where the components $\{x_d, t_d\}$ are the vectors of spatial-temporal variables from the observation dataset $\mathbf{M}_d$; the components $\{x_f, t_f\}$ are the vectors of spatial-temporal variables from a set of collocation points $\mathbf{M}_f$.

The output matrix for the output layer is:

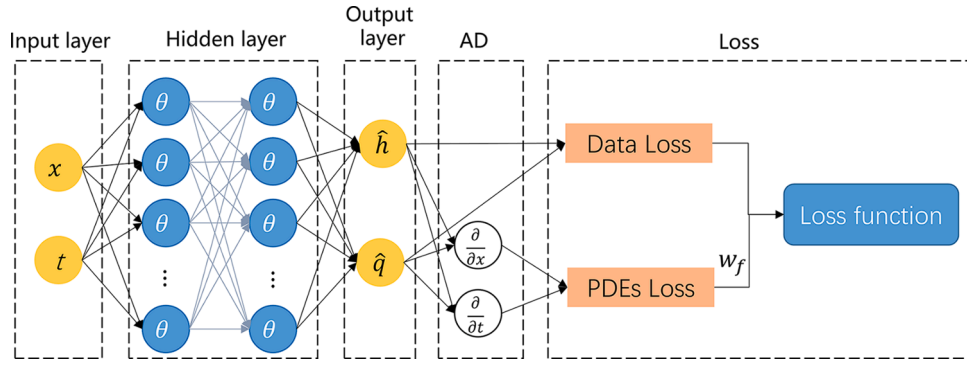


Fig. 2. Schematic diagram of the physics-informed neural networks (PINNs) for the hydraulic transient flow.

$$\text{Output : } \hat{\Phi}(x, t, \theta) = \begin{bmatrix} \hat{h}_d & \hat{q}_d \\ \hat{h}_f & \hat{q}_f \end{bmatrix} = \begin{array}{cc} \text{Head} & \text{Flow} \\ \begin{bmatrix} \hat{h}_d^1 \\ \hat{h}_d^2 \\ \dots \\ \hat{h}_d^{N_d} \\ \hat{h}_f^1 \\ \hat{h}_f^2 \\ \dots \\ \hat{h}_f^{N_f} \end{bmatrix} & \begin{bmatrix} \hat{q}_d^1 \\ \hat{q}_d^2 \\ \dots \\ \hat{q}_d^{N_d} \\ \hat{q}_f^1 \\ \hat{q}_f^2 \\ \dots \\ \hat{q}_f^{N_f} \end{bmatrix} \end{array} \quad (12)$$

To build the PINN model as shown in Fig. 2, a training procedure is required to optimize the neural network parameters θ to minimize the loss function. During the training process, a set of observation points M_d and a set of collocation points M_f are fed to the PINN model. Firstly, the forward propagation is performed in the PINN to compute the output $\hat{\Phi}(x, t, \theta)$, i.e., the head $\hat{h}$ and flow $\hat{q}$, and evaluate the loss, defined by Eqs. (8)–(10). In particular, the head $\hat{h}_d$ and flow $\hat{q}_d$ at the observation points are used to calculate the data loss term L_{data} using Eq. (9). The head $\hat{h}_f$ and flow $\hat{q}_f$ at the collocation points are used to calculate the partial differential terms $\frac{\partial h}{\partial x}$, $\frac{\partial h}{\partial t}$, $\frac{\partial q}{\partial x}$ and $\frac{\partial q}{\partial t}$ through the AD algorithm. These partial differential terms together with $\hat{h}_f$ and $\hat{q}_f$ are subsequently used to calculate the PDEs loss term L_{PDE} specified by Eq. (10). The total loss is then computed by taking the weighted sum as expressed in Eq. (8). Secondly, the gradients of the total loss with respect to the neural network parameters θ are calculated using the back propagation rules. The parameter set θ is iteratively updated through this process on the whole training dataset until the total loss converges. The convergence of the total loss function, in this paper, is evaluated by the Adaptive Moment Estimation (Adam) algorithm (i.e., a variant of stochastic gradient descent (SGD) optimization) with a prescribed stop criterion and by the L-BFGS-B optimizer, a Quasi-Newton based algorithm that can efficiently solve large nonlinear optimization problems (Byrd et al., 1995).

After the training stage, the PINN model can be used to estimate the transient pressure and flowrate traces at any location along the pipeline. In this paper, the performance of the proposed PINN scheme is evaluated on a set of preselected testing points at unmonitored locations. The testing dataset M_e is defined by:

$$M_e = \{x_e^i, t_e^i, \varphi_e^i\}_{i=1}^{N_e} \quad (13)$$

where N_e is the total number of the data points at the selected testing points and at the selected time steps, and the subscript e indicates testing data. The relative L_2 error defined in Eq. (14) is used as the metric for

evaluation.

$$L_2 = \sqrt{\frac{\sum_{i=1}^{N_e} [\hat{\varphi}_e^i(x, t, \theta) - \varphi_e^i(x, t)]^2}{\sum_{i=1}^{N_e} [\varphi_e^i(x, t)]^2}} \quad (14)$$

In addition to these output errors, the pressure traces at testing points are also given to demonstrate the accuracy and efficiency of the proposed method.

3. Numerical studies on reservoir-pipeline-valve system with unknown boundaries

3.1. Reservoir-pipeline-valve system

Fig. 3 depicts the reservoir-pipeline-valve system, which has been widely used to understand and analyze the fundamental transient phenomena occurring in pipeline systems due to its simple form. In this paper, numerical studies on this system were also firstly conducted to validate the proposed PINN-based transient analysis method.

In general, the numerical solution of the transient pressure h and flowrate q along the pipe requires boundary conditions including the information of the transient excitation and the initial condition. For this system, the upstream boundary is often a constant-level reservoir, written as:

$$h_{res}(t) = h_{up} \quad (15)$$

where h_{up} is the upstream water level. At the downstream boundary, the flow is exposed to the atmosphere through a valve. The relationship between the flowrate and pressure at the downstream valve can be written as:

$$q_v(t) - \tau(t)(C_d A_v)_0 \sqrt{2gh_v(t)} = 0 \quad (16)$$

where h_v and q_v is the pressure head and flowrate at the valve, respectively; τ is the relative valve opening and $\tau = 1$ corresponds to the initial valve opening; when C_d is the coefficient of flowrate and A_v is the area of valve opening; the subscript 0 denotes the initial condition.

The transient simulation starts from the steady state where the initial flowrate is constant along the pipe.

3.2. System setup

The reservoir-pipeline-valve system in this case study possesses the following properties: the internal diameter of the pipeline is assumed to be $D = 1.81$ m; the length of the pipeline is $L = 500$ m; the wave speed $a = 1000$ m/s; the Darcy-Weisbach factor $f = 0.012$; the water level of the upstream reservoir $h_{up} = 100$ m; and the initial flowrate q_0 is 0.314 m³/s. To test the capacity of the proposed PINN for analyzing complex wave forms, a continuous valve excitation as shown in Fig. 4 has been applied. In addition, three pressure transducers $\{P_1, P_2, P_3\}$ is assumed to be at

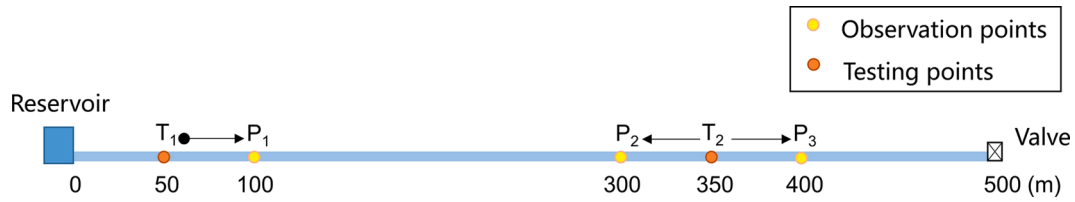


Fig. 3. Pipeline configuration of the numerical case.

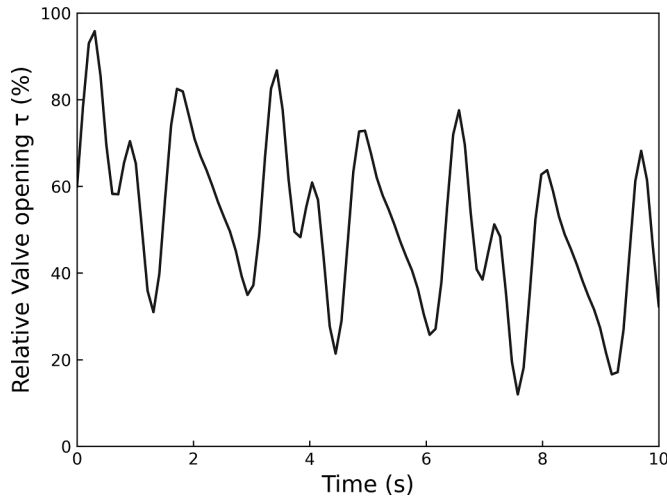


Fig. 4. The relative valve opening of the downstream valve.

100 m, 300 m, and 400 m away from the upstream reservoir as shown in Fig. 3. The transient pressures recorded by these transducers were used to train the PINN model.

Ten seconds of pressure traces at these locations were generated by the method of characteristics (MOC) as shown in Fig. 5. The time step of the pressure from the MOC model was selected as $\Delta T = 0.01$ s. However, the simulated pressure traces were then uniformly downsampled to ΔT

$= 0.05$ s to reduce the computational cost, which results in a total of 200 samples for each transducer. Therefore, the observation data M_d at P_1 , P_2 and P_3 used for the training process contained $N_d = 600$ samples. In addition, 20 uniformly distributed locations were selected as collocation points M_f , resulting in $N_f = 4,000$ collocation samples (20 collocation points with 200 samples at each point). These locations were used to impose the physics of the hydraulic transient to the PINN by minimizing the L_{PDE} term in Eq. (10). Finally, the pressure and flowrate traces at two testing points T_1 ($x = 50$ m) and T_2 ($x = 350$ m) were also generated as the testing dataset M_e for the transient state prediction. The total number of testing points was $N_e = 800$ (two testing points with 200 samples at each point for each of the pressure head and flowrate traces).

It is noted that a reasonable duration of the data series should be determined for specific applications. A long duration can provide more training data for the PINN model, but in the meantime, it increases the complexity of the transient analysis which may require a larger neural network to fit. The selection of the time step mainly depends on the sampling rate of pressure sensors. The higher sampling rate allows the higher resolution of the transient to be captured. Therefore, a small time step is required for simulating a rapid transient, and vice versa.

3.3. PINN for transient analysis in the reservoir-pipeline-valve system

3.3.1. Neural network training

Various PINN structures were evaluated through the training and testing process, in which the following PINN structure obtained the lowest relative L_2 error on the test data: the FNN has 10 layers, including one input layer, one output layer, and eight hidden layers with 20 neurons in each hidden layer. The hyperbolic tangent activation

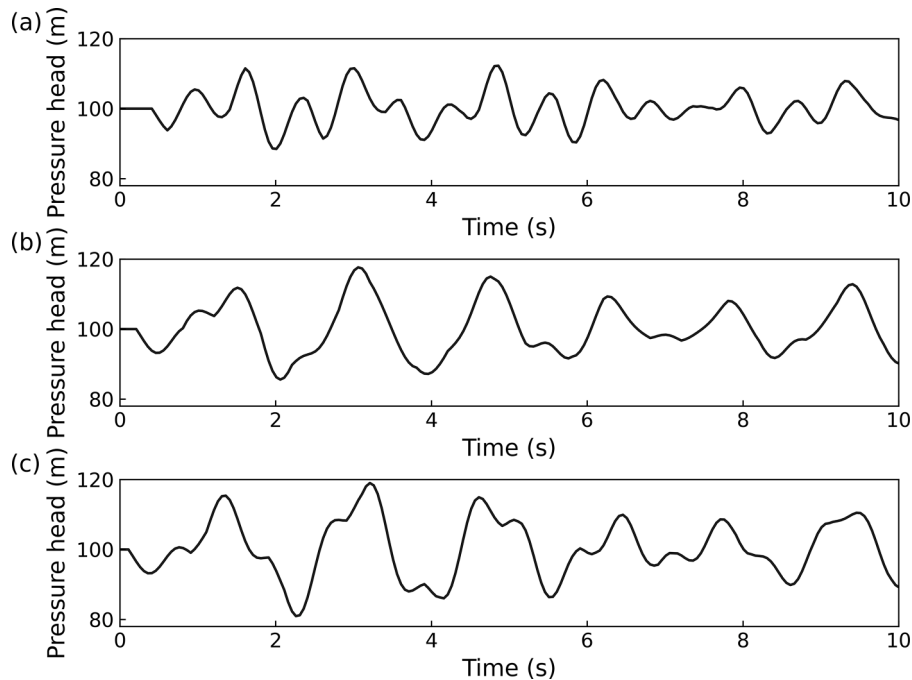


Fig. 5. Observed pressure head traces simulated by MOC (a) at P_1 ($x = 100$ m); (b) at P_2 ($x = 300$ m); (c) P_3 ($x = 400$ m).

function is used for the hidden layers (Tartakovsky et al., 2020) and it is used in all the simulation tests in the present work. The neural network has been trained with the Adam optimizer for 500,000 steps followed by the L-BFGS-B optimizer with 50,000 steps, which took around 48 min on a GPU device of GeForce GTX 1080 Ti. A weight $w_f = 0.1$ in the loss function (Eq. (8)) is determined for PINN through a sensitivity analysis. The details of the sensitivity analysis are reported in Section 3.4. For comparison, a conventional ANN is also built with the same configuration, which can be simply achieved by setting $w_f = 0$ in the constructed PINN.

Fig. 6 plots the convergences of the loss functions of the PINN and the ANN models. It can be seen that the value of the loss function of ANN reached a smaller value in the end. The lower loss values indicate ANN can better fit the training pressure traces. The total loss function $L(\theta)$, the component corresponding to the PDEs loss term L_{PDE} and the component corresponding to the observation data L_{data} were also included in Fig. 6. The stochastic gradient descent method in the Adam algorithm causes the oscillations in the loss function, which allows a better ANN generalization. Then the quasi-Newton L-BFGS-B method enables a higher rate of convergence to the minimum identified by the Adam algorithm.

3.3.2. Pressure and flowrate prediction

The predicted pressure variations at the testing points compared with the simulated results are shown in Fig. 7. The solid lines depict the simulated results by the MOC, the dash lines depict the predicted results by the PINN, and the dot lines depict the predicted results by the ANN. In contrast to the comparison of the total loss in the training process (smaller loss for the ANN and larger loss for the PINN), the relative L_2 error defined in Eq. (14) is 0.4% for the pressure prediction using PINN, while it is 4.78% using ANN. This means the actual prediction (see Figs. 7 and 8) is better when using PINN. Without the constraints of physical laws, the ANN overfits the training data (i.e., pressure traces) at observation points and is not able to fit the testing data at unmonitored locations. This demonstrates the advantage of incorporating physics-related information into the neural network and shows the superiority of the proposed PINN method. This advantage is more obvious when looking at the flowrate prediction results of the ANN and the PINN.

Although the initial boundary (initial flowrate q_0) is not included in the loss function, the PINN is found to be able to predict the variation of flowrate accurately with a different initial value. To match the predicted flowrate variation with the simulated flowrate, the predicted flowrate traces are adjusted by:

$$\tilde{q} = \hat{q} + (q_0 - \hat{q}_0) \quad (17)$$

where $\hat{q}$ is the vector consisting of the originally predicted flowrate trace, $\tilde{q}$ is the vector consisting of the adjusted predicted flowrate trace and q_0 is the predicted initial flowrate (at $t = 0$ s). The predicted flowrate traces after adjustment by Eq. (17) are shown in Fig. 8. The relative L_2 error for the flowrate prediction is 6.1% using PINN, while it is extremely large ($>100\%$) using ANN. The better agreement of flowrate trace using PINN demonstrates its capability of accurate flowrate prediction. Overall, the results illustrate the efficacy of PINN for transient analysis to predict persistent pressure and flow variations at targeted locations along the pipe.

One of the main advantages of PINN is its ability to automatically approximate the partial derivatives in PDEs. This makes the process of solving PDEs and matching the predictions with observations much faster without the need for discretization which is computationally expensive in traditional methods.

Ensuring the correctness of approximated partial derivatives by PINN provides a more rigorous way to validate the proposed PINN-based method. In the transient flow analysis, four different partial derivatives $\frac{\partial h}{\partial x}$, $\frac{\partial h}{\partial t}$, $\frac{\partial q}{\partial x}$, and $\frac{\partial q}{\partial t}$ are constrained by the two governing equations Eqs. (1) and (2) and are embedded in the pressure measurements as well. Those approximated partial derivatives at testing point T_1 were compared with those computed by a numerical method, referred as to four-point-centred finite-difference (FD) scheme (Chaudhry, 2014). With the simulated head and flow by MOC, the partial derivatives of Eqs. (1) and (2) were computed by the FD method described in the Supporting Information.

Fig. 9 shows the comparison of the derivatives at the testing point T_1 between PINN and FD approximations. By substituting the partial derivatives $\frac{\partial h}{\partial x}$, $\frac{\partial h}{\partial t}$, $\frac{\partial q}{\partial x}$, and $\frac{\partial q}{\partial t}$ into Eq. (14) to replace the physical quantity φ , the L_2 error of these predicted derivatives can be obtained as 2.49%, 4.31%, 4.01%, and 2.34%, respectively. The results rigorously confirm that PINN can effectively solve the PDEs to obtain the correct pressure heads and flow rates (in Figs. 7 and 8) and the correct corresponding derivatives (in Fig. 9).

3.4. Sensitivity analysis of the hyperparameters

To balance the trade-off between accuracy and efficiency, it is important to choose appropriate hyperparameters in the training process. Fig. 10 illustrates the effects of the weight on the loss function,

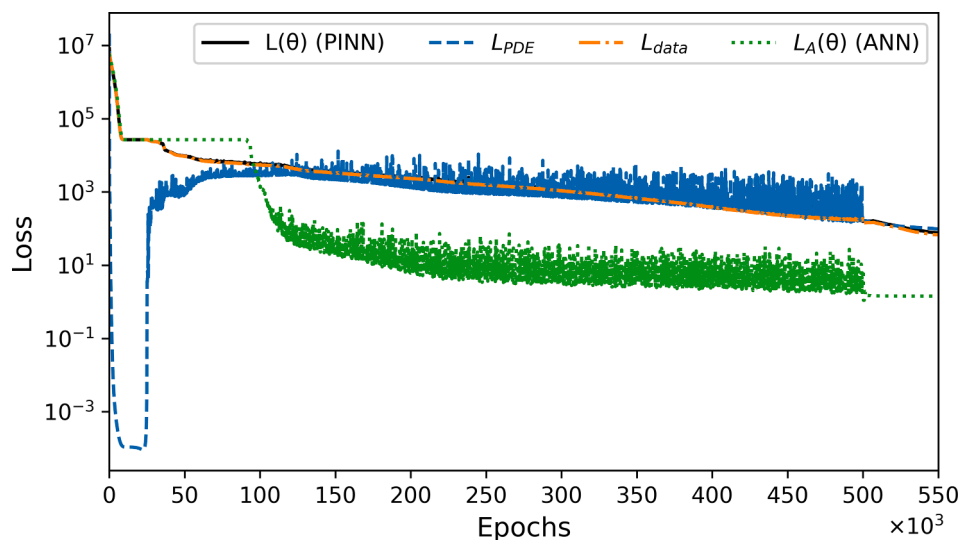


Fig. 6. The loss value $L_A(\theta)$ of conventional ANN compared with the loss values $L(\theta)$, L_{PDE} and L_{data} corresponding to the total loss, PDEs loss and observation loss in the PINN, respectively. 500,000 iterations (epochs) trained with Adam optimizer followed by L-BFGS-B optimizer with the recording interval of 100 iterations.

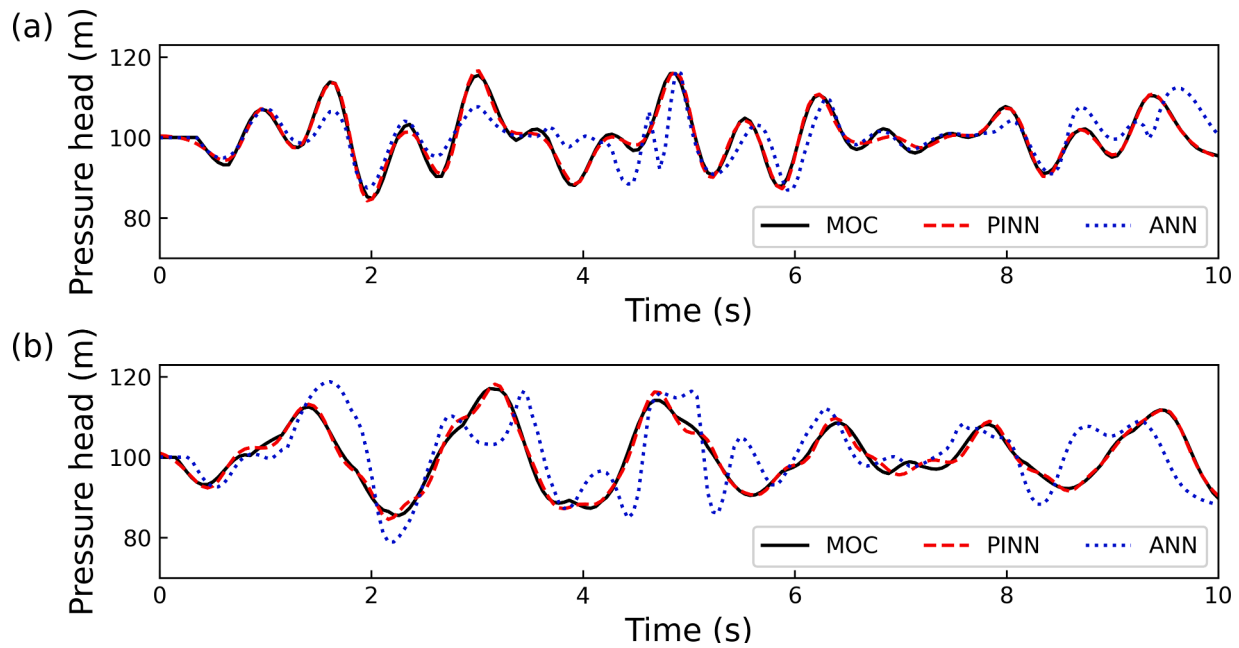


Fig. 7. Predicted and simulated pressure heads at testing points: (a) T_1 ; (b) T_2 .

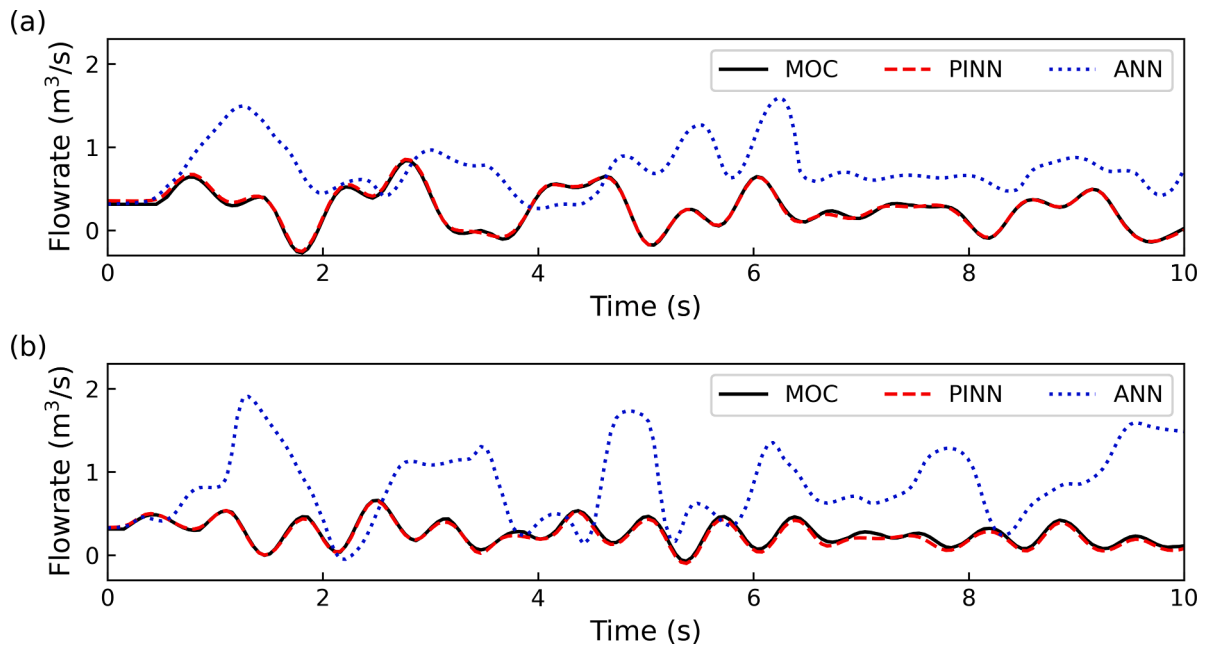


Fig. 8. Predicted and simulated flowrates at testing points: (a) T_1 ; (b) T_2 .

neural network depth (number of layers), neural network width (number of neurons of each hidden layer), and training epochs on the prediction accuracy. To assess the effect of each parameter, the hyperparameters used in the case as shown in Figs. 7 and 8 are set as the benchmark. Only a single parameter varies in each case in the following sensitivity analyses and the remaining parameters keep the same as those of the benchmark.

In Fig. 10(a), when w_f reaches a proper value of 0.1 which penalizes the PDEs loss less than the data loss, the predicted pressure traces match the simulated pressure traces better. Fig. 10(b) indicates that 10 hidden layers are deep enough to achieve good matches of the pressure traces. The result of width = 30 in Fig. 10(c) shows a slightly low relative L_2 error than the result of width = 20. Fig. 10(d) shows that more training

epochs lead to lower relative L_2 error. However, a larger neural network and more training epochs mean increasing computational cost. The training time with epochs of 100 k, 300 k, 500 k, and 800 k in Fig. 10(d) is 10 min, 29 min, 48 min, and 76 min, respectively. A trade-off between accuracy and speed should be made based on the sensitivity analysis. Therefore, the parameters in the benchmark, which achieved satisfactory relative L_2 error = 0.5%, are applied to the PINN architectures in all the numerical experiments in the following.

3.5. Effect of the locations of observation points

In the above case studies, the locations of observation points x_d are selected randomly. However, preliminary studies by the authors show

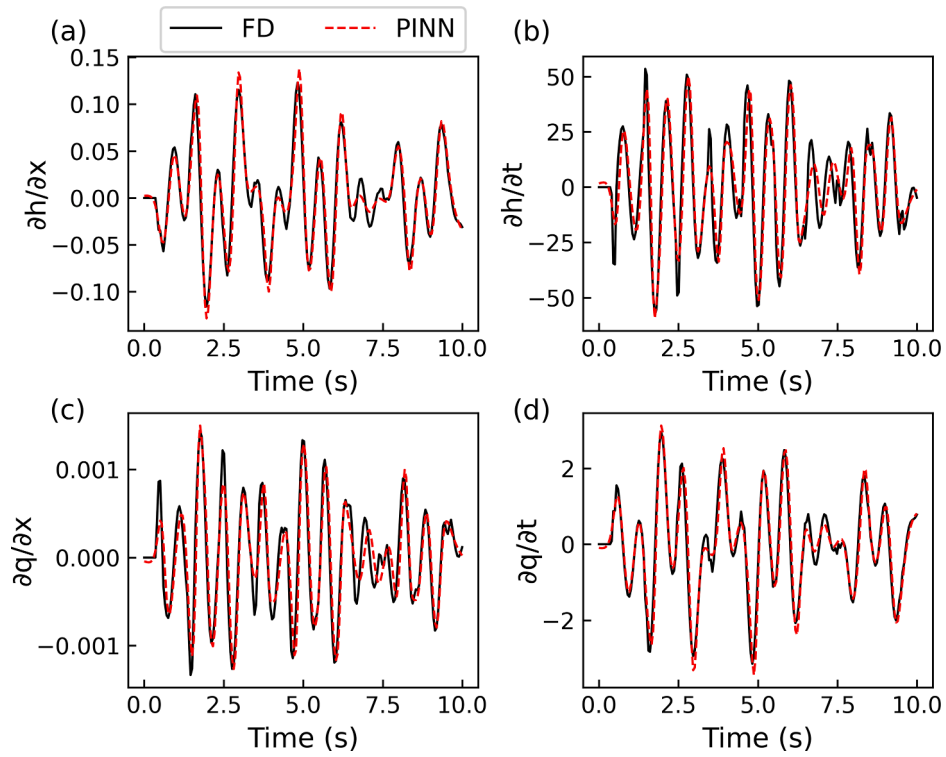


Fig. 9. Comparison of derivatives obtained by PINN and FD at T1: (a) $\frac{\partial h}{\partial x}$; (b) $\frac{\partial h}{\partial t}$; (c) $\frac{\partial q}{\partial x}$; (d) $\frac{\partial q}{\partial t}$.

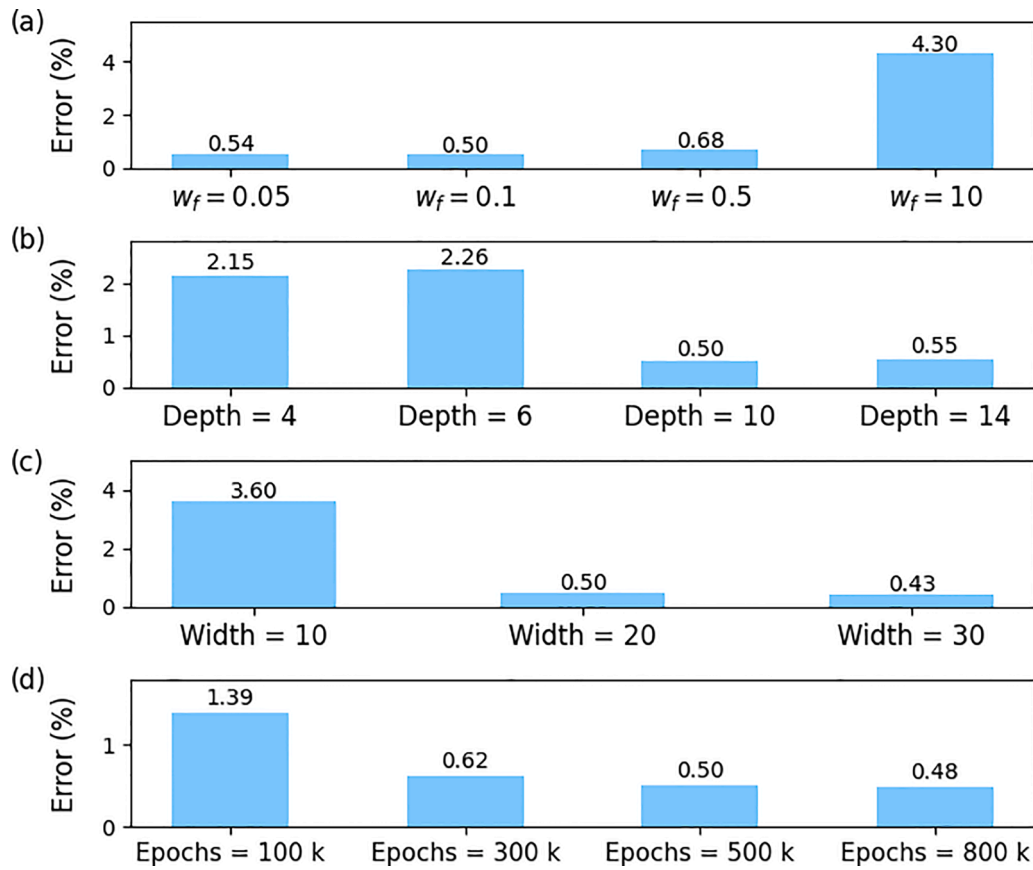


Fig. 10. Relative L_2 error of predicted pressure obtained by PINN with different (a) weights; (b) depth; (c) width; (d) epochs.

the selection of the locations of observation points has a significant effect on the performance of the prediction results. This effect was investigated in this study to explore a better choice of the locations of observation points.

Three different schemes of the locations of three observation points, as shown in Fig. 11, have been investigated to find the optimal sensor placement.

Scheme 1 involves three observation points distributed at 100 m, 200 m and 300 m. Scheme 2 involves two observation points same with Scheme 1 at 100 m and 200 m, respectively and the other one point at 450 m, which is close to the adjacent boundaries. Scheme 3 involves one observation point at the upstream boundary and the other two observation points same as Scheme 2. To make the L_2 error reflect the accuracy of the results generally, 11 testing points are uniformly distributed along the pipe (from 0 m to 500 m).

Fig. 12 presents the relative L_2 errors between simulated pressures obtained by MOC and predicted pressures obtained by the PINN, involving the 11 uniformly distributed testing points and average errors (solid lines) for each scheme. Along the pipe, it can be seen that the testing points near the observation points show lower errors than those far away from the observation points. For example, the testing point at 0 m in Scheme 1 (Fig. 12(a)) or Scheme 2 (Fig. 12(b)) has a larger error than the testing point at 50 m when the nearest observation points are at 100 m and 200 m. However, the errors at 0 m and 50 m in Scheme 2 are smaller than those in Scheme 1, even with the third observation point (at 450 m) in Scheme 2 farther away than the point (at 300 m) in Scheme 1. This indicates that the results are not only affected by the distance between the testing point and the observation point, but also by the distribution of the sensors. When the sensors are placed closer to boundaries that contain essential information about the transient event, the accuracy of the pressure prediction over the whole pipe can be improved. This conclusion is also demonstrated by the results in Scheme 3 which has one observation point exactly at the upstream boundary (0 m) and shows the smallest error of 0.12% between simulated predicted pressures.

4. Numerical studies on reservoir-pipeline-valve system with an incomplete model

4.1. Effect of uncertainties of pipe characteristics

Analyses in the previous section assume the pipe characteristics (e.g., pipe diameter and wave speed) are perfectly known. In this case, the effect of uncertainties of these pipe characteristics on the performance of PINN is investigated by introducing errors to the pipe diameter D or the wave speed a .

The pipe system in Fig. 3 with the same valve excitation in Fig. 4 is considered in this case. The pressure traces at P_1 , P_2 and P_3 simulated by MOC with true pipe parameters ($D = 1.81$ m and $a = 1000$ m/s) are still used as the observation dataset M_d and testing dataset M_e . To introduce parameter errors to the PINN model, the governing equations Eqs. (1) and (2) are set with either incorrect pipe diameter or wave speed. In particular, four different cases have been considered with $D = 1.72$ m, $D = 1.90$ m, $a = 900$ m/s, and $a = 1100$ m/s, respectively. Here, the pipe diameter or wavespeed is changed independently and changing pipe diameter does not change the wave speed. Other parameters are unchanged in these four cases. Fig. 13 shows the simulated pressure variations at T_1 ($x = 50$ m) and T_2 ($x = 350$ m). It can be seen that large deviations are caused by the change in the diameter or wave speed.

Fig. 14 shows the simulated or “true” pressure traces for the benchmark case and the predicted pressures obtained from four PINN models trained with incorrect pipe parameters. The relative L_2 errors for the four cases ($D = 1.72$ m, $D = 1.90$ m, $a = 900$ m/s, and $a = 1100$ m/s) are negligible, with the values of 0.50%, 0.47%, 0.67%, and 0.70%, respectively. It can be seen that the performance of the trained PINN models is consistent and accurate even with consideration of large uncertainties from pipe characteristics.

4.2. Effect of wave damping

4.2.1. Observed data involving wave damping

Factors such as pipe wall viscoelasticity (Covas et al., 2004), unsteady friction (Vardy and Brown, 1995), fluid-structure interaction (Tijsseling, 1996), all contribute to the damping of transient pressure waves. Strong wave dissipation and dispersion have been observed in

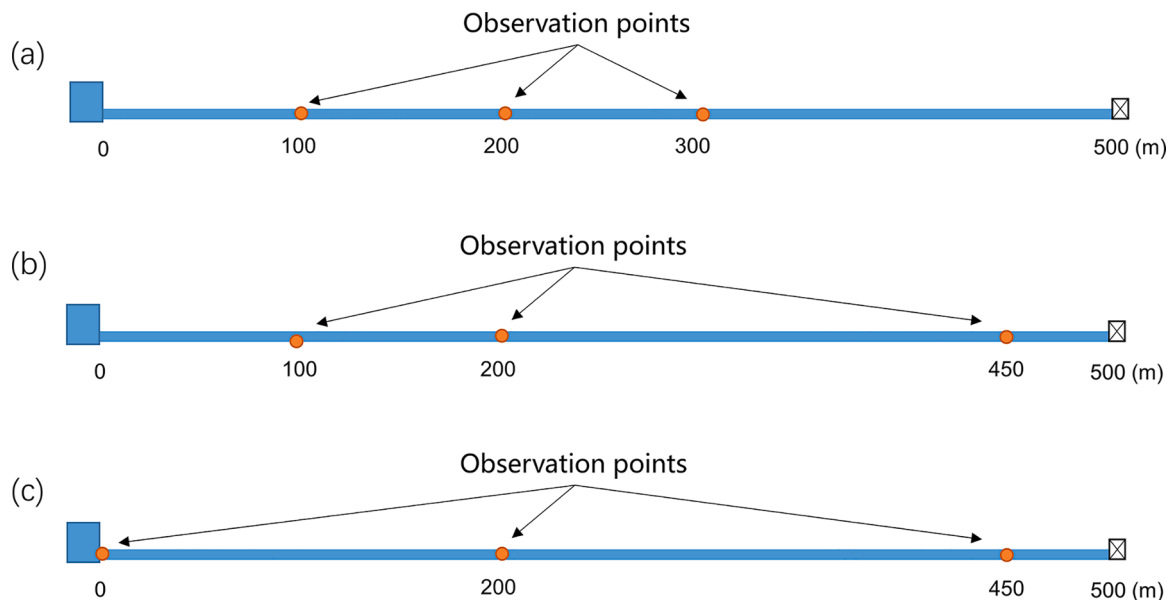


Fig. 11. Schemes of the locations of observed points: (a) Scheme 1 (randomly selected); (b) Scheme 2 (close to the boundaries); (c) Scheme 3 (including upstream boundary)

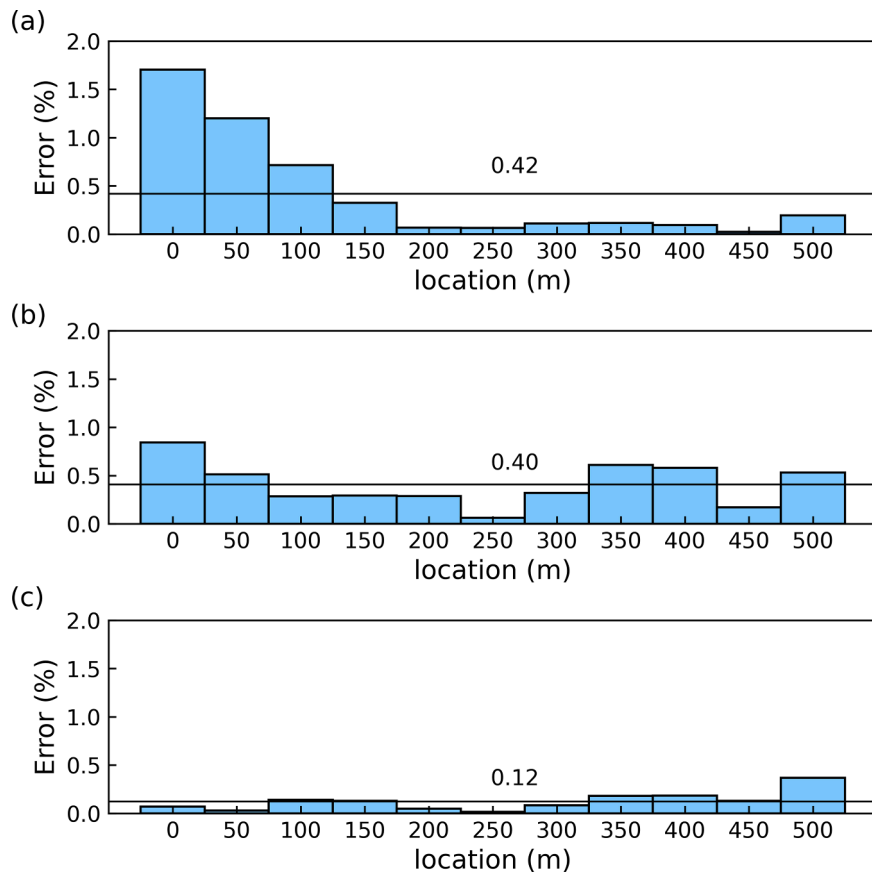


Fig. 12. Relative L_2 errors between simulated pressures obtained by MOC and predicted pressures obtained by the PINN of each testing point and the average errors (solid lines) of (a) Scheme 1; (b) Scheme 2; (3) Scheme 3

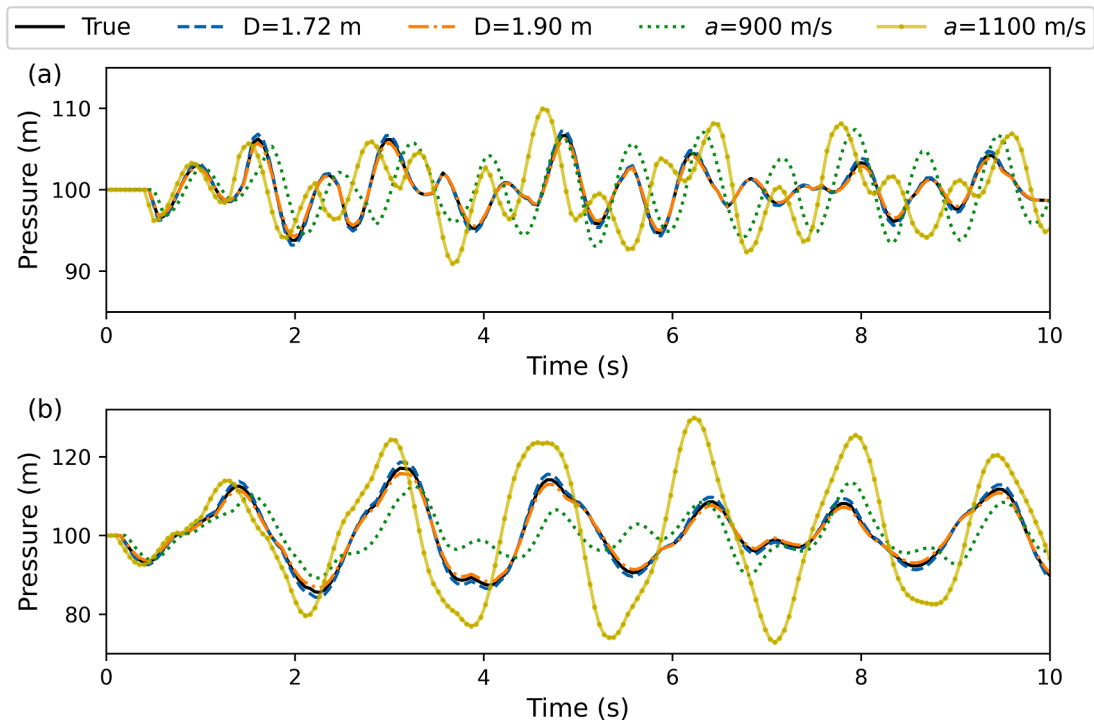


Fig. 13. Effects of the parameter errors on the hydraulic transient pressure simulated by the MOC model (a) at T_1 ($x = 50$ m); (b) at T_2 ($x = 350$ m). The numerical case with true pipe parameters ($D = 1.81$ m and $a = 1000$ m/s) was used to generate the observation and testing datasets.

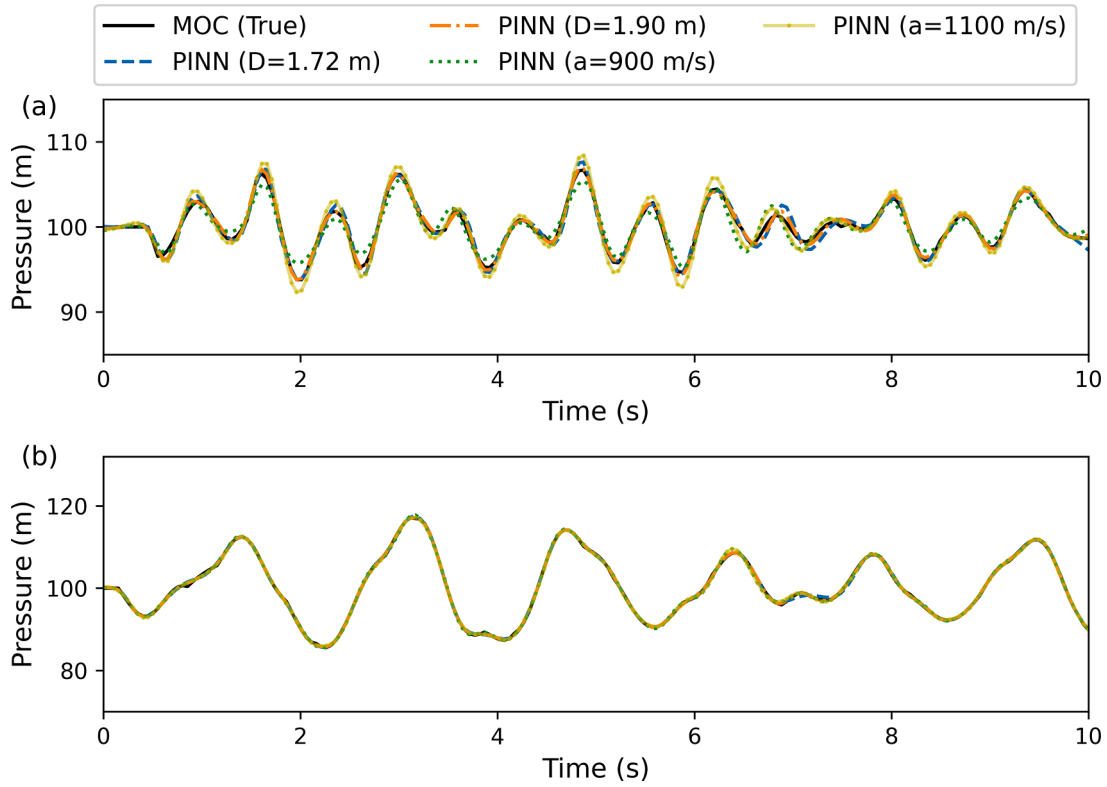


Fig. 14. Simulated pressure heads by MOC with true pipe parameters and predicted pressures by PINNs trained with incorrect pipe parameters (a) at T_1 ($x = 50$ m); (b) at T_2 ($x = 350$ m).

laboratory tests on copper pipes (Budny et al., 1991), high-density polyethylene pipes (Covas et al., 2004) and also on a mild steel cement-lined pipe in the field (Stephens et al., 2011). Therefore, the effect of damping needs to be considered to obtain an accurate characterization of transient flow conditions. However, it is difficult to obtain a well-calibrated physics-based numerical model that can accurately incorporate wave dissipation and dispersion caused by all different factors.

In this section, the observed pressure data involving wave damping were used to train the PINN. However, the effect of damping is not included in the physical laws that guide the training of the PINN, which means the standard PDEs Eqs. (1) and (2) are still used to construct the loss function in Eq. (8) and the viscoelastic model is unknown in the training process.

The following case study shows how the wave damping information embedded in the observed pressure traces is incorporated into the pressure prediction by the PINN with an incomplete physical model.

4.2.2. Dataset generation by a viscoelastic transient model

To generate the training data containing wave dissipation and dispersion, the Kelvin-Voigt viscoelastic model (Covas et al., 2005) has been utilized in this case for hydraulic transient simulation. The viscoelastic model incorporates the dynamic effects of pipe wall viscoelasticity into the hydraulic transient simulation by adding a retarded viscoelastic term in the continuity equation (Eq. (2)), yielding:

$$A \frac{\partial h}{\partial t} + q \frac{\partial h}{\partial x} + \frac{a^2}{g} \frac{\partial q}{\partial x} + \frac{2a^2 A}{g} \frac{\partial \varepsilon_r}{\partial t} = 0 \quad (18)$$

in which ε_r is the retarded strain. The viscoelastic behaviour of the pipe wall is described by a creep compliance function that can be approximated by the following expression:

$$J(t) = J_0 + \sum_{i=1}^N J_i (1 - e^{-t/\tau_i}) \quad (19)$$

in which J_0 = creep-compliance of the first spring defined by $J_0 = 1/E_0$; J_i = creep compliance of the spring of the Kelvin-Voigt element i defined by $J_i = 1/E_i$; E_i = modulus of elasticity of the spring of i -element τ_i = retardation time of the dashpot of i -element, $\tau_i = \mu_i/E_i$ and μ_i is the viscosity of the dashpot of i -element.

Considering the creep function defined by Eq. (19), the term is calculated as the sum of this factor for each Kelvin-Voigt element i :

$$\varepsilon_r(x, t) = \sum_{i=1}^N \varepsilon_{ri}(x, y) = \sum_{i=1}^N \left\{ \frac{\alpha D p g}{2e} \int_0^t [H(x, t - t') - H_0(x)] \frac{J_i}{\tau_i} e^{-t'/\tau_i} dt' \right\} \quad (20)$$

in which is the pipe wall constraint coefficient; e is the wall thickness. The details of the numerical solution of Eqs. (1), (18) and (20) can be found in Covas et al. (2005).

The pipe system shown in Fig. 3 with the same valve excitation as shown in Fig. 4 is used in this case with a one-element viscoelastic model used to model the pipe. The parameters of the pipe are: $J = [5.029 \times 10^{-6}] \text{ Pa}^{-1}$, $\tau = [5.05]$, $\alpha = 1.07$ and $e = 63$ mm and other parameters are the same as those in the previous section. Fig. 15 shows the simulated pressure traces at P_1 , P_2 and P_3 by using the viscoelastic model and the non-viscoelastic model (standard MOC). It can be seen from Fig. 15 that the pressure traces simulated by the viscoelastic model suffer strong damping, which leads to the clear reduction of the magnitude of the pressure waves. The simulated pressures at P_1 , P_2 and P_3 by the viscoelastic model will be used as the observed data in the PINN.

4.2.3. Neural network training and results

Three cases have been conducted with the weight $w_f = 0.1, 0.01$, and 0, respectively. Fig. 16 compares the pressure traces obtained by the

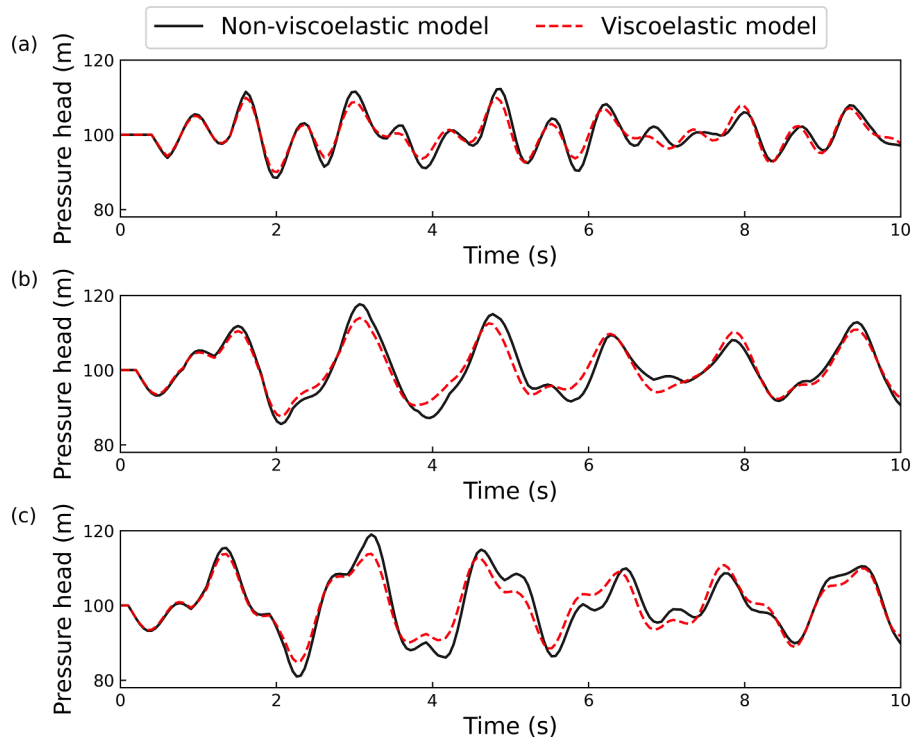


Fig. 15. Observed pressure variations simulated by the viscoelastic model and non-viscoelastic model (standard MOC) (a) at P_1 ($x = 100$ m); (b) at P_2 ($x = 300$ m); (c) P_3 ($x = 400$ m)

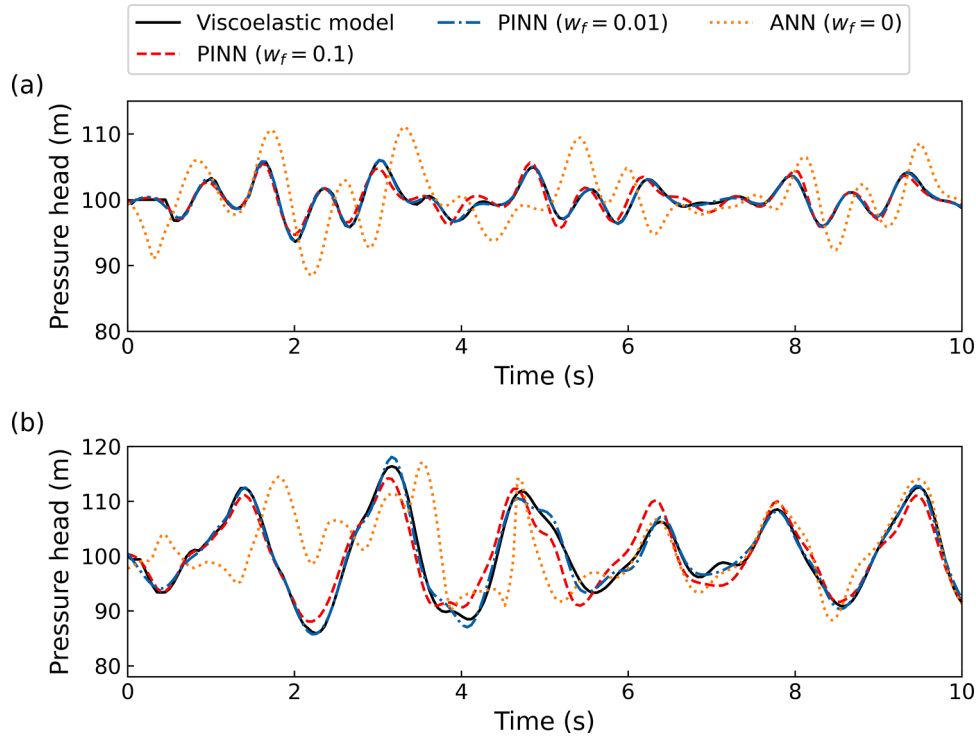


Fig. 16. Simulated pressure heads with wave damping by the viscoelastic model and predicted pressures by PINNs with the weight $w_f = 0.1$, 0.01 , and 0 , respectively: (a) at T_1 ($x = 50$ m); (b) at T_2 ($x = 350$ m).

viscoelastic model and the estimated results by the PINN with three different weights w_f .

In Fig. 16, the results obtained by ANN ($w_f = 0$) showed the largest error as expected. However, compared with the predicted pressure traces with $w_f = 0.1$ which led to the best performance in sensitivity

analysis, the results with $w_f = 0.01$ show much better agreement with the simulated pressure traces. The relative L_2 error with $w_f = 0.01$ is 0.54% , while the relative L_2 error with $w_f = 0.1$ is 1.70% . This indicates that it is necessary to set a smaller w_f to enforce a weaker penalty on the

L_{PDE} term (see Eq. (8)) which is formulated by the incomplete model and inversely a stronger penalty on the L_{data} term. In this way, the observation data containing essential damping information can have dominant effects on the training process, and thus drive the prediction approach to the true damped pressure traces.

Overall, the results in Fig. 16 demonstrate that PINN can successfully incorporate wave damping into transient analysis for pressure prediction even without an accurate physics-based numerical model.

5. Numerical studies on a pipe network

In real practice, most of the pipes are embedded in a pipe network. Transients in a network can be much more complicated due to the complex wave propagation and reflections that depends on the topology of the network. The transient response of the excitation can be formed by a variety of incident and reflected waves, proposing challenges for the PINN to reconstruct the complex transient process.

The applicability and accuracy of the proposed PINN-based method for transient analysis have also been evaluated through numerical transient analysis on a water pipe that is embedded in a pipe network as shown in Fig. 17. This pipe network consists of six nodes (including an upstream reservoir and an end valve) and six pipes. The information about the pipes is given in Table 1. The water level of the upstream reservoir and the initial flowrate through the end valve keep the same as those of the single pipe system (Fig. 3) and are 100 m and 0.314 m³/s, respectively. The valve in the pipe system is assumed to oscillate following the pattern as shown in Fig. 4.

For the pipe network, the PINN-based transient analysis has been carried out on the pipe P_{3-4} with three hypothetical pressure sensors at 50 m, 200 m and 250 m from Node 3, respectively. Assuming the boundary conditions and the transient source are unknown, the PINN-based method is used to predict pressure variations at some selected locations on the pipe P_{3-4} .

The transient process of the whole pipe network has been stimulated by the MOC model with a time step ΔT of 0.01 s and a total time $T = 10$ s. The simulated pressure traces at the three sensors and their corresponding locations and time series are used to form the observation dataset M_d (Eq. (4)). Ten uniformly distributed locations on the pipe P_{3-4} are selected for the collocation dataset M_f (Eq. (7)).

The PINN is built with the same layers and hyperparameters as that for the reservoir-pipeline-valve system. To test the applicability of the PINN-based method in this pipe network, the predicted pressure traces at Node 3 and Node 4 are represented in Fig. 18. The relative L_2 errors are 2.69% for Node 3 and 1.39% for Node 4. The predicted pressure trace at Node 4 shows better agreement with the simulation results. This is because there are two sensors near Node 4 but only one sensor near Node 3. The good fitness between predicted and simulated pressure traces demonstrates the ability of the proposed PINN-based method for transient analysis on a pipe that is embedded in a network with unknown boundary conditions.

Table 1

Pipe parameters for the example pipe network.

Pipe	Start node	End node	Length (m)	Diameter (m)	Wave speed (m/s)	Friction factor
P_{1-2}	1	2	100	1.81	1000	0.012
P_{2-3}	2	3	100	0.56	1000	0.012
P_{3-4}	3	4	300	0.56	1000	0.012
P_{4-5}	4	5	100	0.56	1000	0.012
P_{5-6}	5	6	100	1.81	1000	0.012
P_{2-5}	2	5	300	1.81	1000	0.012

6. Experimental verification

Laboratory experiments have been conducted on a single copper pipeline system in the Robin Hydraulics Laboratory at the University of Adelaide to validate the PINN-based method for hydraulic transient analysis.

Fig. 19 shows the layout of the experimental pipeline system. A reservoir-pipeline-valve system was configured by connecting the pipeline with a pressurized tank at the upstream side, and a dead-end was created by the closure of the in-line valve at the downstream side. The pipeline is 37.21 m in length with an internal diameter D of 22.14 mm throughout the pipe. The wave speed of the pressure wave in the pipe was calculated using the theoretical formula as $a = 1319$ m/s. The Darcy–Weisbach friction factor is assumed as 0.012. Five pressure transducers are installed along the pipe with a sampling time step Δt of 0.0001 s. The locations of these pressure transducers can be found in Fig. 19. A side-discharge solenoid valve was located 144 mm downstream from the closed inline valve, for the generation of transient waves.

In the transient experiment, the side-discharge solenoid valve was closed suddenly. A period of 1 s, in which the pressure wave can propagate back and forth a couple of times, was selected for the analysis. The measured pressure traces by the transducers are shown in Fig. 20. Several kinds of uncertainties are included in the measured data, such as unsteady friction, damping, sensor noise, system errors, and other uncertainties associated with the experiments.

To reduce the computational cost, the pressure traces were uniformly downsampled to 0.001 s. In this case, the weight w_f in Eq. (8) equals 50, which was determined by a sensitivity analysis. The architecture and training steps of the neural network are the same as before.

Three cases are considered here by regarding different transducer locations as observation points. Table 2 summarizes the two cases with their relative L_2 errors given. The results demonstrate a good pressure prediction by PINN as all the relative L_2 errors in the cases are within 2%. Case 2 shows a slightly lower error than Case 1 because of more observation points.

Figs. 21 and 22 present the experimental and predicted pressures at the testing points for each case. It can be seen that the proposed PINN-based method can well predict the pressure trace whether the testing point is between the observation points (Point 3) or out of them (Point 5).

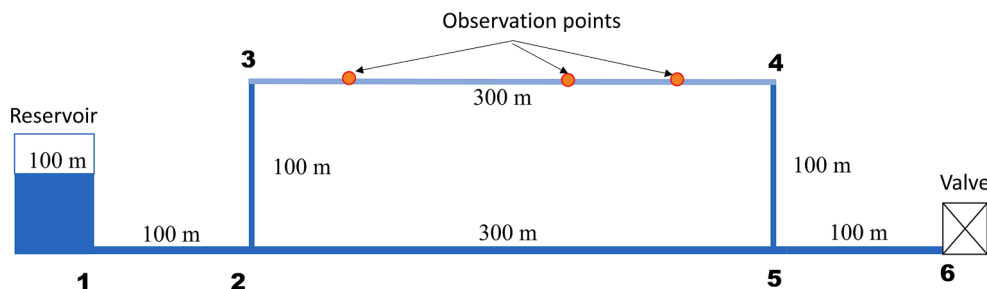


Fig. 17. The pipe network configuration of the numerical case.

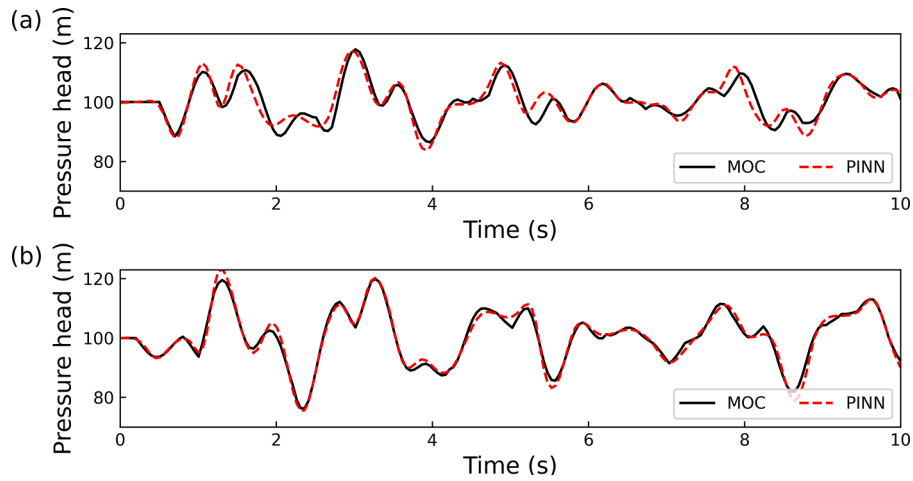


Fig. 18. Simulated (MOC) and predicted (PINN) pressure variations: (a) at Node 3; (b) at Node 4 in the pipe network.

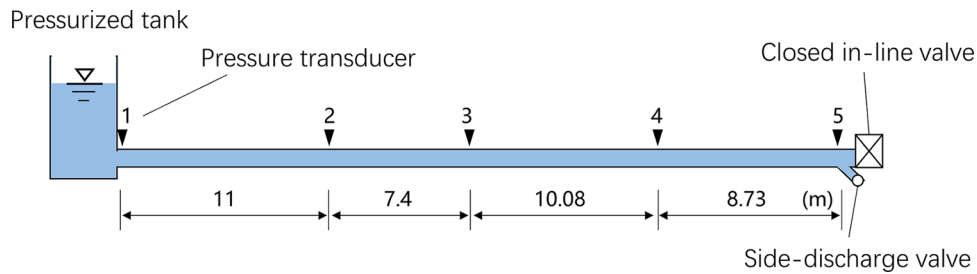


Fig. 19. Experimental pipeline system.

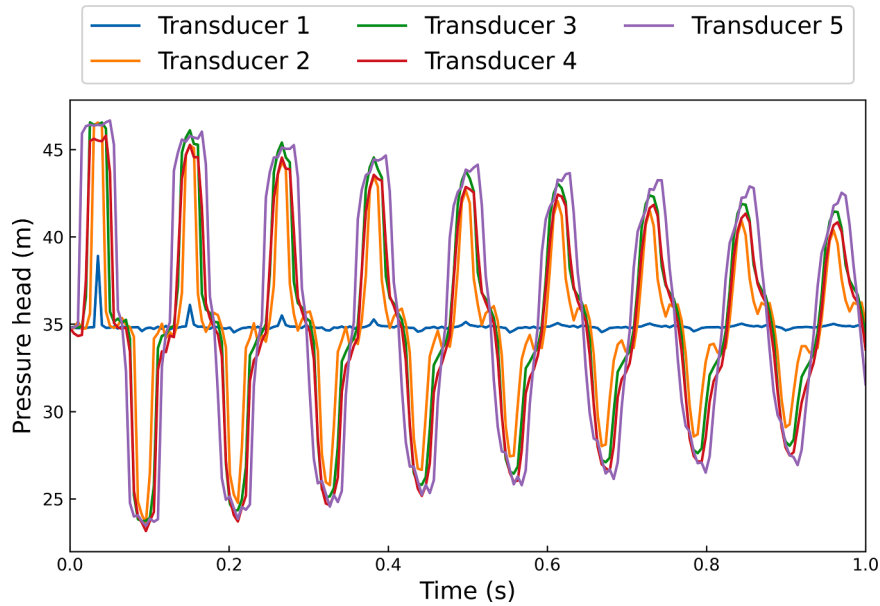


Fig. 20. Measured pressures in the transient experiment.

Table 2

Two cases and their corresponding relative L_2 errors.

Cases	Observation points	Testing points	Relative L_2 error
Case 1	2, 4	3, 5	1.3%
Case 2	1, 2, 4	3, 5	0.8%

7. Conclusions

In this paper, a data-driven PINN-based method has been proposed for one-dimensional hydraulic transient analysis. The proposed method is capable of predicting transient pressure (and flowrate) variations at unmonitored locations with limited measurements and without the requirement of a complete physical model of the pipe system (for example, boundary conditions and/or the wave propagation

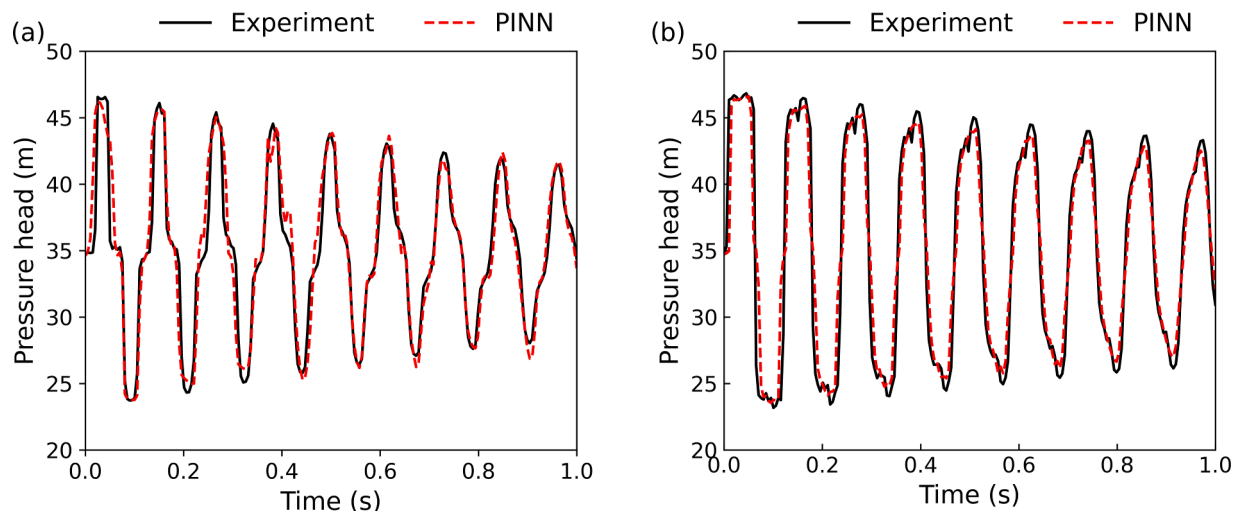


Fig. 21. Experimental and predicted pressures: (a) at Testing Point 3; (b) at Testing Point 5 of Case 1.

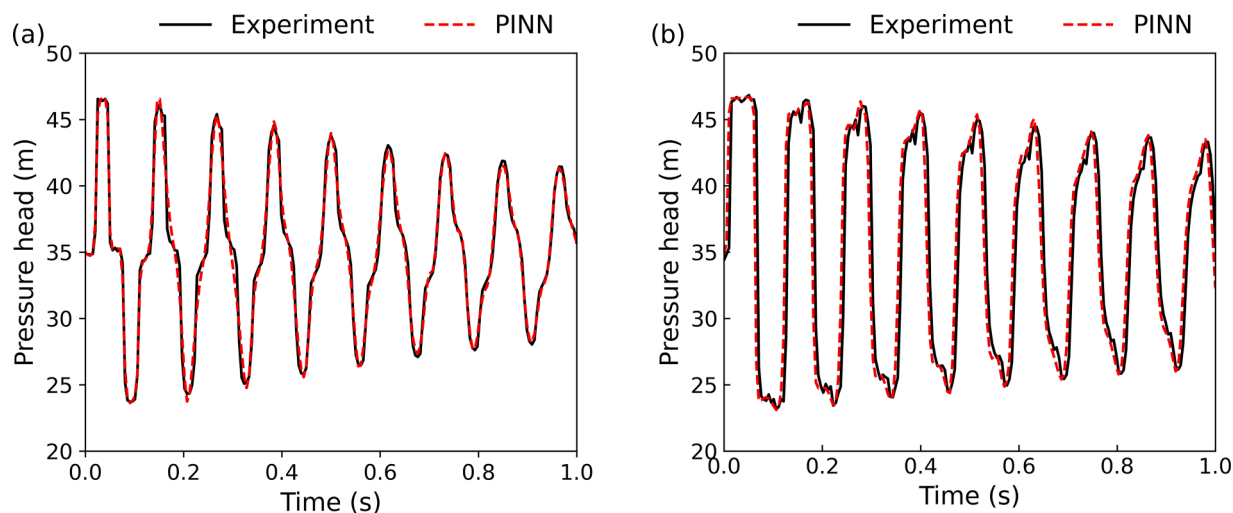


Fig. 22. Experimental and predicted pressures: (a) at Testing Point 3; (b) at Testing Point 5 of Case 2.

characteristics can be unknown). The new method can incorporate the missing physical information that is embedded in the measured data into the transient analysis constrained by the physical laws (unsteady continuity and momentum equations). Numerical and experimental studies were carried out to validate the applicability and accuracy of this method. The key findings are shown as follows.

The numerical case in a reservoir-pipe-valve system shows the proposed PINN-based method can accurately predict pressure and flowrate variations with unknown boundary conditions. The method has a much better performance than the traditional ANN which ignores the key governing equations in the prediction. Further sensitivity analyses on this pipe system show the locations and number of the observation points in the learning process can have an effect on the predicted results. Putting the sensors closer to the boundary is found to be able to effectively improve the accuracy of the pressure prediction. The numerical case using observation data incorporating wave damping shows that the PINN-based transient analysis is reliable even with an incomplete model which misses the physical representation of the wave damping in governing equations (e.g. creep function of the viscoelastic behaviour of the pipe wall in this case). Apart from the reservoir-pipe-valve system, the proposed PINN-based method is also validated on a pipe that is embedded in a network which is a more common situation in real practice. In the end, laboratory experiments with high accuracy show

the potential of this method to be applied to real WDSs.

The above results showcase the potential for successful application of the PINN-based method in real transmission pipes, unlocking a series of opportunities for water supply security and optimization, achieving good accuracy and high computational speed given various uncertainties in system parameters, boundary conditions and physical processes. As this is the first application of transient flow and PINNs, the proposed approach has only been evaluated on simple pipeline systems and for a single transient scenario. The extension of this research will focus on applications of the PINN approach to more complex pipe networks. Furthermore, the development of the PINN model for arbitrary transient events will be considered in future work.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.watres.2022.118828](https://doi.org/10.1016/j.watres.2022.118828).

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